

Table 4.3 Tail aspect ratio and taper ratio

	Horizontal tail		Vertical tail	
	A	λ	A	λ
Fighter	3-4	0.2-0.4	0.6-1.4	0.2-0.4
Sail plane	6-10	0.3-0.5	1.5-2.0	0.4-0.6
Others	3-5	0.3-0.6	1.3-2.0	0.3-0.6
T-Tail	—	—	0.7-1.2	0.6-1.0

taper ratio. Note that T-tail aircraft have lower vertical-tail aspect ratios to reduce the weight impact of the horizontal tail's location on top of the vertical tail. Also, some general-aviation aircraft use untapered horizontal tails ($\lambda = 1.0$) to reduce manufacturing costs.

Leading-edge sweep of the horizontal tail is usually set to about 5 deg more than the wing sweep. This tends to make the tail stall after the wing, and also provides the tail with a higher Critical Mach Number than the wing, which avoids loss of elevator effectiveness due to shock formation. For low-speed aircraft, the horizontal tail sweep is frequently set to provide a straight hinge line for the elevator, which usually has the left and right sides connected to reduce flutter tendencies.

Vertical-tail sweep varies between about 35 and 55 deg. For a low-speed aircraft, there is little reason for vertical-tail sweep beyond about 20 deg other than aesthetics. For a high-speed aircraft, vertical-tail sweep is used primarily to insure that the tail's Critical Mach Number is higher than the wing's.

The exact planform of the tail surfaces is actually not very critical in the early stages of the design process. The tail geometries are revised during later analytical and wind-tunnel studies. For conceptual design, it is usually acceptable simply to draw tail surfaces that "look right," based upon prior experience and similar designs.

Tail thickness ratio is usually similar to the wing thickness ratio, as determined by the historical guidelines provided in the wing-geometry section. For a high-speed aircraft, the horizontal tail is frequently about 10% thinner than the wing to ensure that the tail has a higher Critical Mach Number.

Note that a lifting canard or tandem wing should be designed using the guidelines and procedures given for initial wing design, instead of the tail-design guidelines described above.

THRUST-TO-WEIGHT RATIO AND WING LOADING

5.1 INTRODUCTION

The thrust-to-weight ratio (T/W) and the wing loading (W/S) are the two most important parameters affecting aircraft performance. Optimization of these parameters forms a major part of the analytical design activities conducted after an initial design layout. The methods used for this optimization are described in Chapter 19.

However, it is essential that a credible estimate of the wing loading and thrust-to-weight ratio be made before the initial design layout is begun. Otherwise, the optimized aircraft may be so unlike the as-drawn aircraft that the design must be completely redone.

For example, if the wing loading used for the initial layout is very low, implying a large wing, the designer will have no trouble finding room for the landing gear and fuel tanks. If later optimization indicates the need for a much higher wing loading, the resulting smaller wing may no longer hold the landing gear and fuel. While they could be put in the fuselage, this would increase the wetted area and therefore the drag, so the optimization results would probably be no good.

Wing loading and thrust-to-weight ratio are interconnected for a number of performance calculations, such as takeoff distance, which is frequently a critical design driver. A requirement for short takeoff can be met by using a large wing (low W/S) with a relatively small engine (low T/W). While the small engine will cause the aircraft to accelerate slowly, it only needs to reach a moderate speed to lift off the ground.

On the other hand, the same takeoff distance could be met with a small wing (high W/S) provided that a large engine (high T/W) is also used. In this case, the aircraft must reach a high speed to lift off, but the large engine can rapidly accelerate the aircraft to that speed.

Due to this interconnection, it is frequently difficult to use historical data to independently select initial values for wing loading and thrust-to-weight ratio. Instead, the designer must guess at one of the parameters and use that guess to calculate the other parameter from the critical design requirements.

In many cases, the critical requirement for wing loading will be the stall speed during the approach for landing. Approach stall speed is independent of engine size, so the wing loading can be estimated based upon stall speed alone. The estimated wing loading can then be used to calculate the T/W required to attain other performance drivers such as the single-engine rate of climb.

For less obvious cases, the designer must guess one parameter, calculate the other parameter to meet various performance requirements, then recheck the first parameter. In this book, the thrust-to-weight ratio appears as the first guess because that parameter better lends itself to a statistical approach, and also because it shows less variation within a given class of aircraft.

However, for certain aircraft the designer may wish to begin instead with the wing loading. In such cases, those equations presented below for calculating the wing loading can be solved for thrust-to-weight ratio instead.

5.2 THRUST-TO-WEIGHT RATIO

Thrust-to-Weight Definitions

T/W directly affects the performance of the aircraft. An aircraft with a higher T/W will accelerate more quickly, climb more rapidly, reach a higher maximum speed, and sustain higher turn rates. On the other hand, the larger engines will consume more fuel throughout the mission, which will drive up the aircraft's takeoff gross weight to perform the design mission.

T/W is not a constant. The weight of the aircraft varies during flight as fuel is burned. Also, the engine's thrust varies with altitude and velocity (as does the horsepower and propeller efficiency, η_p).

When designers speak of an aircraft's thrust-to-weight ratio they generally refer to the T/W during sea-level static (zero-velocity), standard-day conditions at design takeoff weight and maximum throttle setting. Another commonly referred-to T/W concerns combat conditions.

You can also calculate T/W at a partial-power setting. For example, during the approach to landing the throttle setting is near idle. The operating T/W at that point in the mission is probably less than 0.05.

It is very important to avoid confusing the takeoff T/W with the T/W at other conditions in the calculations below. If a required T/W is calculated at some other condition, it must be adjusted back to takeoff conditions for use in selecting the number and size of the engines. These T/W adjustments will be discussed later.

Power Loading and Horsepower-to-Weight

The term "thrust-to-weight" is associated with jet-engined aircraft. For propeller-powered aircraft, the equivalent term has classically been the "power loading," expressed as the weight of the aircraft divided by its horsepower (W/hp).

Power loading has an opposite connotation from T/W because a high power loading indicates a smaller engine. Power loadings typically range from 10–15 for most aircraft. An aerobatic aircraft may have a power loading of about six. A few aircraft have been built with power loadings as low as three or four. One such over-powered airplane was the Pitts Sampson, a one-of-a-kind airshow airplane.

A propeller-powered aircraft produces thrust via the propeller, which has an efficiency η_p defined as the thrust output per horsepower provided by the

engine. Using Eq. (3.9), an equivalent T/W for propellered aircraft can therefore be expressed as follows:

$$\frac{T}{W} = \left(\frac{550 \eta_p}{V} \right) \left(\frac{\text{hp}}{W} \right) \quad (5.1)$$

Note that this equation includes the term hp/W , the horsepower-to-weight ratio. This is simply the inverse of the classical power loading (W/hp). To avoid confusion when discussing requirements affecting both jet- and propeller-powered aircraft, this book refers to the horsepower-to-weight ratio rather than the classical power loading. The reader should remember that the power loading can be determined simply by inverting the horsepower-to-weight ratio.

Also, to avoid excessive verbiage in the discussions below, the term "thrust-to-weight ratio" should be understood to include the horsepower-to-weight ratio for propeller aircraft.

Statistical Estimation of T/W

Tables 5.1 and 5.2 provide typical values for T/W and hp/W for different classes of aircraft. Table 5.2 also provides reciprocal values, i.e., power loadings, for propellered aircraft. These values are all at maximum power settings at sea level and zero velocity ("static").

Table 5.1 Thrust-to-weight ratio (T/W)

Aircraft type	Typical installed T/W
Jet trainer	0.4
Jet fighter (dogfighter)	0.9
Jet fighter (other)	0.6
Military cargo/bomber	0.25
Jet transport	0.25

Table 5.2 Horsepower-to-weight ratio

Aircraft type	Typical hp/W	Typical power loading (W/hp)
Powered sailplane	0.04	25
Homebuilt	0.08	12
General aviation—single engine	0.07	14
General aviation—twin engine	0.17	6
Agricultural	0.09	11
Twin turboprop	0.20	5
Flying boat	0.10	10

Note that the current generation of dogfighters approaches a T/W of 1.0, implying that the thrust is nearly equal to the weight. At combat conditions when some fuel has been burned off, these aircraft have T/W values exceeding 1, and are capable of accelerating while going straight up! The jet dogfighter T/W values are with afterburning engines, whereas the other jets typically do not have afterburning.

Thrust-to-weight ratio is closely related to maximum speed. Later in the design process, aerodynamic calculations of drag at the design maximum speed will be used, with other criteria, to establish the required T/W .

For now, Tables 5.3 and 5.4 provide curve-fit equations based upon maximum Mach number or velocity for different classes of aircraft. These can be used as a first estimate for T/W or hp/W . The equations were developed by the author using data from Ref. 1, and should be considered valid only within the normal range of maximum speeds for each aircraft class.

Thrust Matching

For aircraft designed primarily for efficiency during cruise, a better initial estimate of the required T/W can be obtained by "thrust matching." This refers to the comparison of the selected engine's thrust available during cruise to the estimated aircraft drag.

In level unaccelerating flight, the thrust must equal the drag. Likewise, the weight must equal the lift (assuming that the thrust is aligned with the

flight path). Thus, T/W must equal the inverse of L/D [Eq. (5.2)]:

$$\left(\frac{T}{W}\right)_{\text{cruise}} = \frac{1}{(L/D)_{\text{cruise}}} \quad (5.2)$$

L/D can be estimated in a variety of ways. Chapter 12 will discuss the detailed drag-buildup approach. For the first estimation of T/W the method for L/D estimation presented in Chapter 3 is adequate.

Recall that this procedure for L/D estimation uses the selected aspect ratio and an estimated wetted-area ratio (Fig. 3.5) to determine the wetted aspect ratio. Figure 3.6 is then used to estimate the maximum L/D . For propeller aircraft, the cruise L/D is the same as the maximum L/D . For jet aircraft, the cruise L/D is 86.6% of the maximum L/D .

Note that this method assumes that the aircraft is cruising at approximately the optimum altitude for the as-yet-unknown wing loading. The method would be invalid if the aircraft were forced by the mission requirements to cruise at some other altitude, such as sea level.

When the wing loading has been selected, as described later in this chapter, the L/D at the actual cruise conditions should be calculated and used to recheck the initial estimate for T/W .

The thrust-to-weight ratio estimated using Eq. (5.2) is at cruise conditions, not takeoff. The aircraft will have burned off part of its fuel before beginning the cruise, and will burn off more as the cruise progresses.

Also, the thrust of the selected engine will be different at the cruise conditions than at sea-level, static conditions. These factors must be considered to arrive at the required takeoff T/W , used to size the engine.

The highest weight during cruise occurs at the beginning of the cruise. The weight of the aircraft at the beginning of the cruise is the takeoff weight less the fuel burned during takeoff and climb to cruise altitude. From Table 3.2, the typical mission weight fractions for these mission legs are 0.970 and 0.985, or 0.956 when multiplied together.

A typical aircraft will therefore have a weight at the beginning of cruise of about 0.956 times the takeoff weight. This value is used below to adjust the cruise T/W back to takeoff conditions.

Thrust during cruise is different from the takeoff value. Jet aircraft are normally designed to cruise at approximately the altitude at which the selected engine has the best (lowest) specific fuel consumption, typically 30,000–40,000 ft. While SFC is improved at these altitudes, the thrust decreases. Also, the engine is sized using the thrust setting that produces the best SFC. This is usually 70–100% of the maximum continuous, nonafterburning thrust.

The cruise thrust at altitude is therefore less than the maximum takeoff thrust at sea-level, so the required cruise T/W must be adjusted to obtain the equivalent takeoff T/W .

Typically, a subsonic, high-bypass-ratio turbofan for a transport aircraft will have a cruise thrust of 20–25% of the takeoff thrust, while a low-bypass afterburning turbofan or turbojet will have a cruise thrust of 40–70% of the

Table 5.3 T/W_0 vs $M_{\max}$

$T/W_0 = a M_{\max}^C$	a	C
Jet trainer	0.488	0.728
Jet fighter (dogfighter)	0.648	0.594
Jet fighter (other)	0.514	0.141
Military cargo/bomber	0.244	0.341
Jet transport	0.267	0.363

Table 5.4 hp/W_0 vs $V_{\max}$ (mph)

$hp/W_0 = a V_{\max}^C$	a	C
Sailplane—powered	0.043	0
Homebuilt—metal/wood	0.005	0.57
Homebuilt—composite	0.004	0.57
General aviation—single engine	0.024	0.22
General aviation—twin engine	0.034	0.32
Agricultural aircraft	0.008	0.50
Twin turboprop	0.012	0.50
Flying boat	0.029	0.23

takeoff maximum value (see Fig. 5.1). Appendix A.4 provides thrust and fuel-consumption data for several representative engines.

For a piston-powered, propeller-driven aircraft, the power available varies with the density of the air provided to the intake manifold. If the engine is not supercharged, then the power falls off with increasing altitude according to the density ratio, σ . For example, a nonsupercharged engine at 10,000 ft will have about 73% of its sea-level power.

To prevent this power decrease, many piston engines use a supercharger to maintain the air provided to the manifold at essentially sea-level density up to the compression limit of the supercharger. Above this altitude, the power begins to drop off (see Fig. 5.2). Piston-powered aircraft typically cruise at about 75% of takeoff power.

For a turbine-powered, propeller-driven (turboprop) aircraft, the horsepower available increases somewhat with increasing speed, but the thrust drops off anyway due to the velocity effect on the propeller [Eq. (5.1)].

With a turboprop, there is an additional, residual thrust contribution from the turbine exhaust. It is customary to convert this thrust to its horsepower equivalent and add it to the actual horsepower, creating an "equivalent shaft horsepower (eshp)." For a typical turboprop engine installation, the cruise eshp is about 60–80% of the takeoff value.

The takeoff T/W required for cruise matching can now be approximated using Eq. (5.3). The ratio between initial cruise weight and takeoff weight was shown to be about 0.956. If a better estimate of this ratio is available, it should be used.

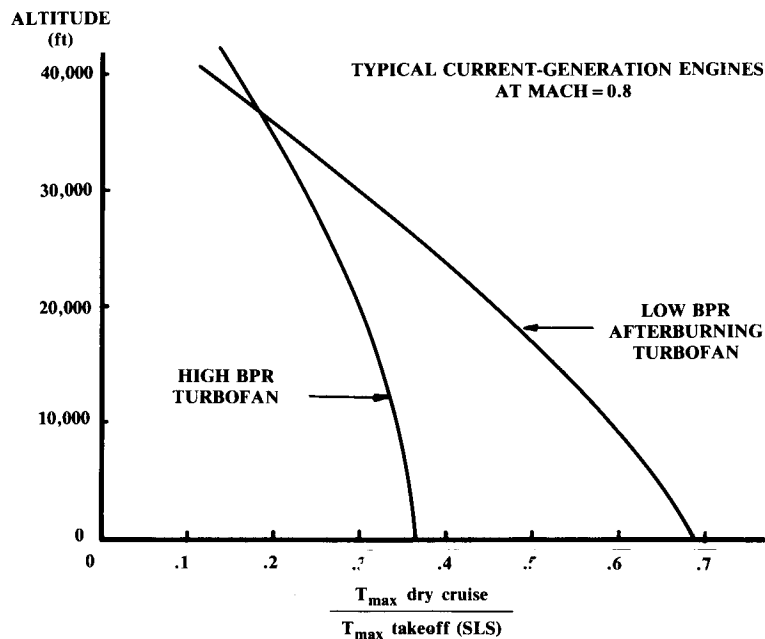


Fig. 5.1 Thrust lapse at cruise.

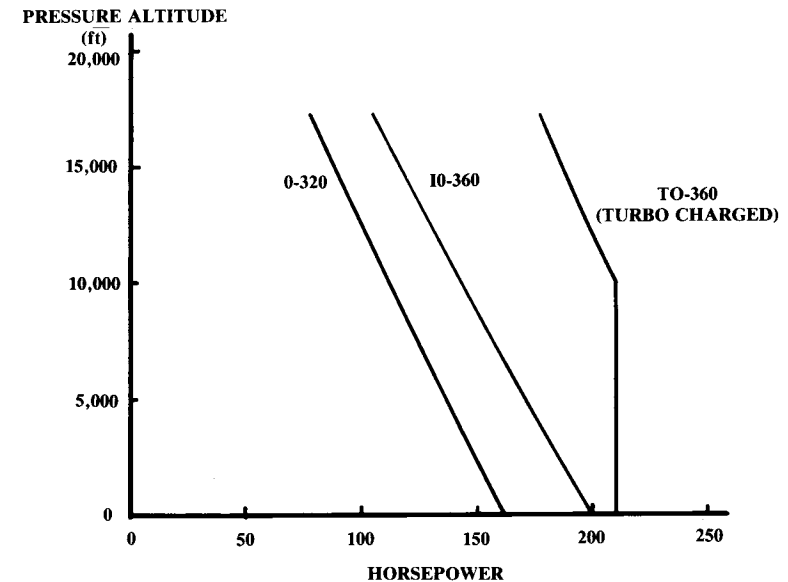


Fig. 5.2 Piston engine power variation with altitude.

$$\left(\frac{T}{W}\right)_{\text{takeoff}} = \left(\frac{T}{W}\right)_{\text{cruise}} \left(\frac{W_{\text{cruise}}}{W_{\text{takeoff}}}\right) \left(\frac{T_{\text{takeoff}}}{T_{\text{cruise}}}\right) \quad (5.3)$$

The thrust ratio between takeoff and cruise conditions should be obtained from actual engine data if possible. Otherwise, data for a similar engine from Appendix A.4 or some other source should be used.

For a propeller aircraft, the required takeoff hp/ W can be found by combining Eqs. (5.1) and (5.2):

$$\left(\frac{\text{hp}}{W}\right)_{\text{takeoff}} = \left(\frac{V_{\text{cruise}}}{550 \eta_p}\right) \left(\frac{1}{(L/D)_{\text{cruise}}}\right) \left(\frac{W_{\text{cruise}}}{W_{\text{takeoff}}}\right) \left(\frac{\text{hp}_{\text{takeoff}}}{\text{hp}_{\text{cruise}}}\right) \quad (5.4)$$

where typically $\eta_p = 0.8$.

After an initial layout has been completed, actual aerodynamic calculations are made to compare the drag during cruise with the thrust available.

There are many other criteria which can set the thrust-to-weight ratio, such as climb rate, takeoff distance, and turning performance. These other criteria also involve the wing loading and are described in the next section.

For the first-pass estimate, the T/W (or hp/W) should be selected as the higher of either the statistical value obtained from the appropriate equation in Tables 5.3 and 5.4, or the value obtained from the cruise matching as described above. After selection of the wing loading as described below, the selected T/W should be rechecked against all requirements.

Table 5.5 Wing loading

Historical trends	Typical takeoff W/S (lb/ft ²)
Sailplane	6
Homebuilt	11
General aviation—single engine	17
General aviation—twin engine	26
Twin turboprop	40
Jet trainer	50
Jet fighter	70
Jet transport/bomber	120

5.3 WING LOADING

The wing loading is the weight of the aircraft divided by the area of the reference (not exposed) wing. As with the thrust-to-weight ratio, the term “wing loading” normally refers to the takeoff wing loading, but can also refer to combat and other flight conditions.

Wing loading affects stall speed, climb rate, takeoff and landing distances, and turn performance. The wing loading determines the design lift coefficient, and impacts drag through its effect upon wetted area and wing span.

Wing loading has a strong effect upon sized aircraft takeoff gross weight. If the wing loading is reduced, the wing is larger. This may improve performance, but the additional drag and empty weight due to the larger wing will increase takeoff gross weight to perform the mission. The leverage effect of the sizing equation will require a more-than-proportional weight increase when factors such as drag and empty weight are increased.

Table 5.5 provides representative wing loadings.

Wing loading and thrust-to-weight ratio must be optimized together. Such optimization methods are presented in Chapter 19 using aerodynamic, weight, and propulsion data calculated from the initial design layout. The remainder of this chapter provides methods for initially estimating the wing loading to meet various requirements. These allow the designer to begin the layout with some assurance that the design will not require a complete revision after the aircraft is analyzed and sized.

This material generally assumes that an initial estimate of T/W has been made using the methods presented in the last section. However, most of the equations could also be used to solve for T/W if the wing loading is defined by some unique requirement (such as stall speed).

These methods estimate the wing loading required for various performance conditions. To ensure that the wing provides enough lift in all circumstances, the designer should select the lowest of the estimated wing loadings. However, if an unreasonably low wing loading value is driven by only one of these performance conditions, the designer should consider another way to meet that condition.

For example, if the wing loading required to meet a stall speed requirement is well below all other requirements, it may be better to equip the aircraft with a high-lift flap system. If takeoff distance or rate of climb require a very low wing loading, perhaps the thrust-to-weight ratio should be increased.

Stall Speed

The stall speed of an aircraft is directly determined by the wing loading and the maximum lift coefficient. Stall speed is a major contributor to flying safety, with a substantial number of fatal accidents each year due to “failure to maintain flying speed.” Also, the approach speed, which is the most important factor in landing distance and also contributes to post-touchdown accidents, is defined by the stall speed.

Civil and military design specifications establish maximum allowable stall speeds for various classes of aircraft. In some cases the stall speed is explicitly stated. FAR 23 certified aircraft (under 12,500-lb TOGW) must stall at no more than 61 knots, unless they are multiengine and meet certain climb requirements (see Appendix A.5). While not stated in any design specifications, a stall speed of about 50 knots would be considered the upper limit for a civilian trainer or other aircraft to be operated by low-time pilots.

The approach speed is required to be a certain multiple of the stall speed. For civil applications, the approach speed must be at least 1.3 times the stall speed. For military applications, the multiple must be at least 1.2 (1.15 for carrier-based aircraft). Approach speed may be explicitly stated in the design requirements or will be selected based upon prior, similar aircraft. Then the required stall speed is found by division by 1.3, 1.2, or 1.15.

Equation (5.5) states that lift equals weight in level flight, and that at stall speed, the aircraft is at maximum lift coefficient. Equation (5.6) solves for the required wing loading to attain a given stall speed with a certain maximum lift coefficient. The air density, ρ , is typically the sea-level standard value (0.00238 slugs/cubic ft) or sometimes the 5000-ft-altitude, hot-day value (0.00189) to ensure that the airplane can be flown into Denver during summer.

$$W = L = q_{\text{stall}} S C_{L_{\text{max}}} = \frac{1}{2} \rho V_{\text{stall}}^2 S C_{L_{\text{max}}} \quad (5.5)$$

$$W/S = \frac{1}{2} \rho V_{\text{stall}}^2 C_{L_{\text{max}}} \quad (5.6)$$

The remaining unknown, the maximum lift coefficient, can be very difficult to estimate. Values range from about 1.2 to 1.5 for a plain wing with no flaps to as much as 5.0 for a wing with large flaps immersed in the propwash or jetwash.

The maximum lift coefficient for an aircraft designed for short takeoff and landing (STOL) applications will typically be about 3.0. For a regular transport aircraft with flaps and slats (leading-edge flaps with slots to improve airflow), the maximum lift coefficient is about 2.4. Other aircraft, with flaps on the inner part of the wing, will reach a lift coefficient of about 1.6–2.0.

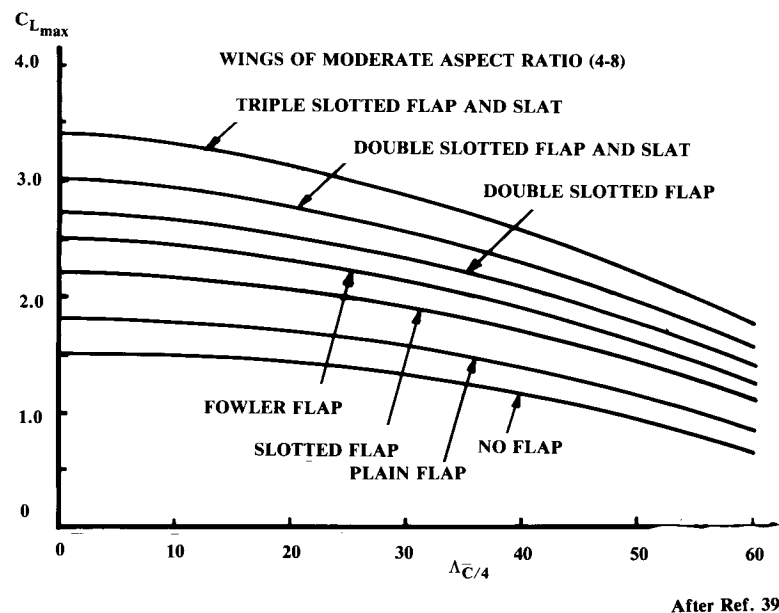


Fig. 5.3 Maximum lift coefficient.

Maximum lift coefficient depends upon the wing geometry, airfoil shape, flap geometry and span, leading-edge slot or slat geometry, Reynolds number, surface texture, and interference from other parts of the aircraft such as the fuselage, nacelles, or pylons. The trim force provided by the horizontal tail will increase or reduce the maximum lift, depending upon the direction of the trim force. If the propwash or jetwash impinges upon the wing or the flaps, it will also have a major influence upon maximum lift during power-on conditions.

Most aircraft use a different flap setting for takeoff and landing. During landing, the flaps will be deployed the maximum amount to provide the greatest lift and drag. However, for takeoff the maximum flap angle will probably cause more drag than desirable for rapid acceleration and climb, so the flaps will be deployed to about half the maximum angle. Therefore, the maximum lift coefficient for landing will be greater than for takeoff. Typically, the takeoff maximum lift coefficient is about 80% that of the landing value.

For a wing of fairly high aspect ratio (over about 5), the maximum lift coefficient will be approximately 90% of the airfoil maximum lift coefficient at the same Reynolds number, provided that the lift distribution is nearly elliptical. However, if partial-span flaps are used, their deflection will introduce a large, discontinuous twist into the wing geometry that changes the lift distribution, and thus the induced downwash, causing the effective angle of attack to vary at different span stations.

As a crude approximation, the designer can ignore this effect. Then the maximum lift can be estimated by determining the maximum angle of attack before some part of the wing stalls. Typically, the part of the wing with the flap deflected will stall first. Then, for that angle of attack the lift contributions of the flapped and unflapped sections can be summed, weighted by their areas (see Fig. 12.19 for definitions of flapped and unflapped areas). This crude approximation for wings of a fairly high aspect ratio is given in Eq. (5.7).

$$C_{L_{\max}} \cong 0.9 \left\{ (C_{l_{\max}})_{\text{flapped}} \frac{S_{\text{flapped}}}{S_{\text{ref}}} + (C_l)_{\text{unflapped}} \frac{S_{\text{unflapped}}}{S_{\text{ref}}} \right\} \quad (5.7)$$

where $C_{l_{\text{unflapped}}}$ is the lift coefficient of the unflapped airfoil at the angle of attack at which the flapped airfoil stalls.

For a better initial estimate of maximum lift, it is necessary to resort to test results and historical data. Figure 5.3 provides maximum-lift trends vs sweep angle for several classes of aircraft. Remember that the maximum lift using the takeoff flap setting will typically be about 80% of these landing maximum values. Maximum lift is discussed in more detail in Chapter 12.

Takeoff Distance

A number of different values are referred to as "takeoff distance." The "ground roll" is the actual distance traveled before the wheels leave the ground. The liftoff speed for a normal takeoff is 1.1 times the stall speed.

The "obstacle clearance distance" is the distance required from brake release until the aircraft has reached some specified altitude. This is usually 50 ft for military or small civil aircraft and 35 ft for commercial aircraft.

The "balanced field length" is the length of the field required for safety in the event of an engine failure at the worst possible time in a multiengine aircraft. When the aircraft has just begun its ground roll, the pilot would have no trouble stopping it safely if one engine were to fail. As the speed increases, more distance would be required to stop after an engine failure. If the aircraft is nearly at liftoff speed and an engine fails, the pilot would be unable to stop safely and instead would continue the takeoff on the remaining engines.

The speed at which the distance to stop after an engine failure exactly equals the distance to continue the takeoff on the remaining engines is called the "decision speed." The balanced field length is the length required to takeoff and clear the specified obstacle when one engine fails exactly at decision speed. Note that use of reversed thrust is not permitted for calculation of balanced field length.

Balanced field length is sometimes called the "FAR takeoff field length" when applied to FAR 25 aircraft (which have a 35-ft obstacle clearance requirement). For military aircraft the balanced field length has a 50-ft obstacle clearance requirement.

Both the wing loading and the thrust-to-weight ratio contribute to the takeoff distance. The equations below assume that the thrust-to-weight ratio has been selected and can be used to determine the required wing loading to attain some required takeoff distance. However, the equations could be solved for T/W if the wing loading is known.

Other factors contributing to the takeoff distance are the aircraft's aerodynamic drag and rolling resistance. Aerodynamic drag on the ground depends largely upon pilot technique. For example, if the pilot rotates (lifts the nose) too early, the extra drag may prevent the aircraft from accelerating to takeoff speed. This was a frequent cause of accidents in early jets, which were underpowered by today's standards.

The aircraft's rolling resistance, μ , is determined by the type of runway surface and by the type, number, inflation pressure, and arrangement of the tires. A thin, high-pressure tire operated over a soft dirt runway will have so much rolling resistance that the aircraft may be unable to move. A large, low pressure tire can operate over a softer runway surface but will have more aerodynamic drag if not retracted, or will take up more room if retracted. Values of μ for different runway surfaces are provided in the detailed takeoff analysis in Chapter 17.

In later stages of analysis the takeoff distance will be calculated by integrating the accelerations throughout the takeoff, considering the variations in thrust, rolling resistance, weight, drag, and lift. For initial estimation of the required wing loading, a statistical approach for estimation of takeoff distance can be used.

Figure 5.4, based upon data from Refs. 4 and 5, permits estimation of the takeoff ground roll, takeoff distance to clear a 50-ft obstacle, and FAR

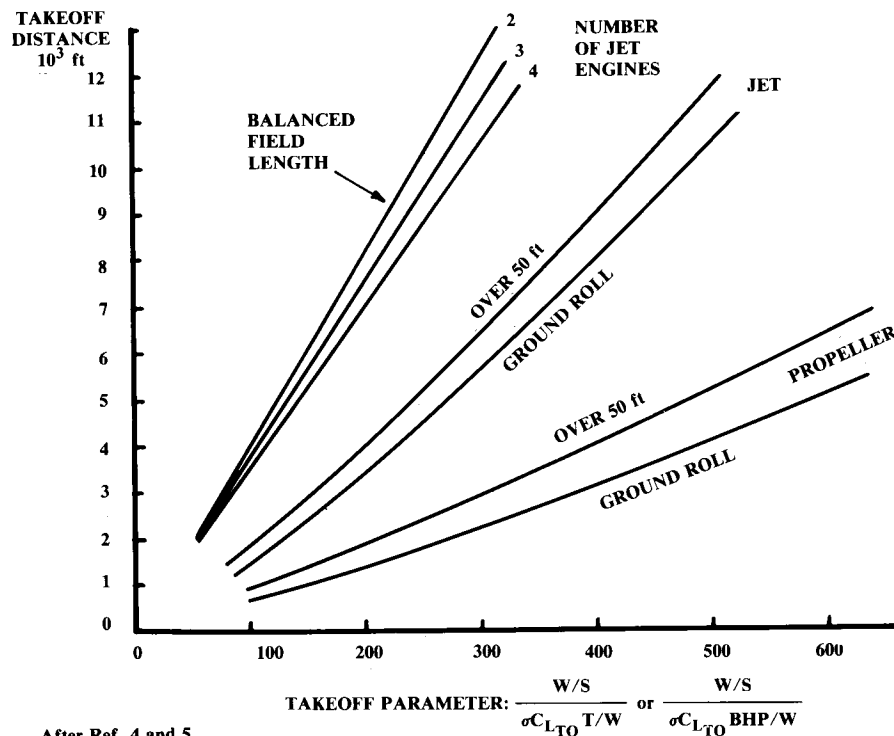


Fig. 5.4 Takeoff distance estimation.

balanced field length over a 35-ft obstacle. For a military multiengine aircraft, the balanced field length over a 50-ft obstacle is approximately 5% greater than the FAR (35-ft) balanced field value.

Note that a twin-engine aircraft has a greater FAR balanced field length than a three- or four-engine aircraft with the same total thrust. This occurs because the twin-engine aircraft loses half its thrust from a single engine failure, whereas the three- and four-engine aircraft lose a smaller percentage of their total thrust from a single engine failure.

The takeoff parameter (TOP) of Fig. 5.4 is the takeoff wing loading divided by the product of the density ratio (σ), takeoff lift coefficient, and takeoff thrust-to-weight (or horsepower-to-weight) ratio. The density ratio is simply the air density (ρ) at the takeoff altitude divided by the sea level density (0.00238 slugs/cubic ft).

The takeoff lift coefficient is the actual lift coefficient at takeoff, not the maximum lift coefficient at takeoff conditions as used for stall calculation. The aircraft takes off at about 1.1 times the stall speed so the takeoff lift coefficient is the maximum takeoff lift coefficient divided by 1.21 (1.1 squared). However, takeoff (and landing) lift coefficient may also be limited by the maximum tail down angle permitted by the landing gear (typically not more than 15 deg).

To determine the required wing loading to meet a given takeoff distance requirement, the takeoff parameter is obtained from Fig. 5.4. Then the following expressions give the maximum allowable wing loading for the given takeoff distance:

$$\text{Prop: } (W/S) = (\text{TOP}) \sigma C_{L_{TO}} (\text{hp}/W) \quad (5.8)$$

$$\text{Jet: } (W/S) = (\text{TOP}) \sigma C_{L_{TO}} (T/W) \quad (5.9)$$

Catapult Takeoff

Most Naval aircraft must be capable of operation from an aircraft carrier. For takeoff from a carrier, a catapult accelerates the aircraft to flying speed in a very short distance.

Catapults are steam-operated, and can produce a maximum force on the aircraft depending on the steam pressure used. Therefore, a light aircraft can be accelerated to a higher speed by the catapult than a heavy one. Figure 5.5 depicts the velocities attainable as a function of aircraft weight for three catapults in use by the U.S. Navy. Note that a rough guess of takeoff weight is required.

For a catapult takeoff, the airspeed as the aircraft leaves the catapult must exceed the stall speed by 10%. Airspeed is the sum of the catapult end speed (V_{end}) and the wind-over-deck of the carrier (V_{wod}).

For aircraft launch operations the carrier will be turned into the wind which will produce a wind-over-deck on the order of 20–40 kt. However, the design specifications for a Navy aircraft frequently require launch capabilities with zero wind-over-deck or even a negative value, to enable aircraft launch while at anchor. Once the end speed is known, the maximum wing loading is defined by:

$$\left(\frac{W}{S}\right)_{\text{landing}} = \frac{1}{2} \rho (V_{\text{end}} + V_{\text{wod}})^2 \frac{(C_{L_{\text{max}}})_{\text{takeoff}}}{1.21} \quad (5.10)$$

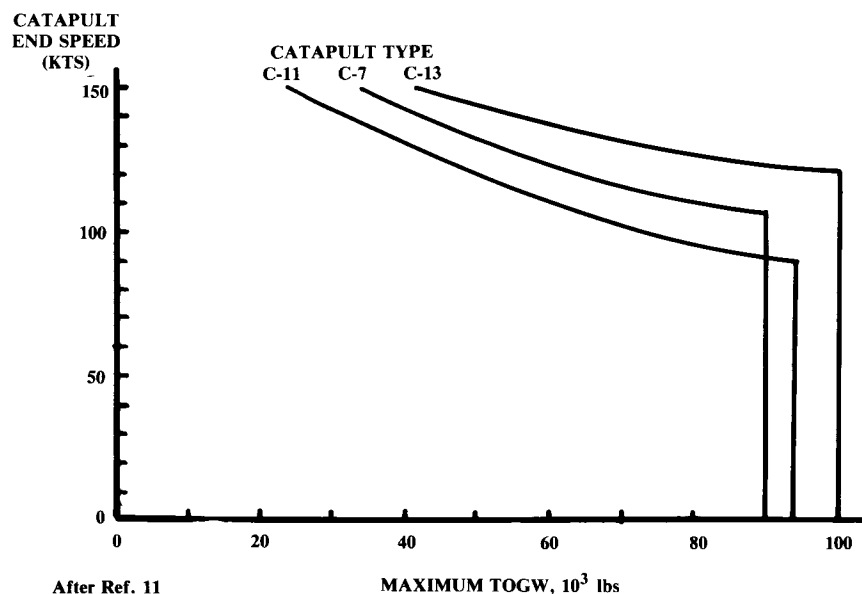


Fig. 5.5 Catapult end speeds.

Landing Distance

There are a number of different values referred to as the "landing distance." "Landing ground roll" is the actual distance the aircraft travels from the time the wheels first touch to the time the aircraft comes to a complete stop.

The "FAR 23 landing field length" includes clearing a 50-ft obstacle while the aircraft is still at approach speed and on the approach glidepath (normally 3 deg). After crossing the obstacle, the pilot slows the aircraft to the touchdown speed of typically 1.15 times the stall speed. The obstacle-clearance distance roughly doubles the ground-roll distance alone.

The "FAR 25 landing field length" includes the 50-ft obstacle clearance at approach speed, and also adds an arbitrary two-thirds to the total distance to allow a safety margin. The landing distance definition for military aircraft is normally specified in a request for proposals (RFP), but typically resembles the FAR 23 definition.

Landing distance is largely determined by wing loading. Wing loading affects the approach speed, which must be a certain multiple of stall speed (1.3 for civil aircraft, 1.2 for military aircraft). Approach speed determines the touchdown speed, which in turn defines the kinetic energy which must be dissipated to bring the aircraft to a halt. The kinetic energy, and hence the stopping distance, varies as the square of the touchdown speed.

In fact, a reasonable first-guess of the total landing distance in feet, including obstacle clearance, is approximately 0.3 times the square of the

approach speed in knots (Ref. 5). This is approximately true for FAR 23 and military aircraft without thrust reversers, and FAR 25 aircraft with thrust reversers. While the FAR 25 aircraft have the additional requirement of a two-thirds distance increase, the thrust reversers used on most FAR 25 aircraft shorten the landing distance by about the same amount.

Equation (5.11) provides a better approximation of the landing distance, which can be used to estimate the maximum landing wing loading. The first term represents the ground roll to absorb the kinetic energy at touchdown speed. The constant term, S_a , represents the obstacle-clearance distance.

$$S_{\text{landing}} = 80 \left(\frac{W}{S} \right) \left(\frac{1}{\sigma C_{L_{\text{max}}}} \right) + S_a \quad (5.11)$$

where

$$\begin{aligned} S_a &= 1000 \text{ (airliner-type, 3-deg glideslope)} \\ &= 600 \text{ (general aviation-type power-off approach)} \\ &= 450 \text{ (STOL, 7-deg glideslope)} \end{aligned}$$

For aircraft equipped with thrust reversers or reversible-pitch propellers, multiply the ground portion of the landing [first term in Eq. (5.11)] by 0.66.

For commercial (FAR 25) aircraft, multiply the total landing distance calculated with Eq. (5.11) by 1.67 to provide the required safety margin.

The landing wing loading must be converted to takeoff conditions by dividing by the ratio of landing weight to takeoff weight. This ratio is usually not based upon the calculated end-of-mission weight, but is instead based upon some arbitrary landing weight as specified in the design requirements.

For most propeller-powered aircraft and jet trainers, the aircraft must meet its landing requirement at or near the design takeoff weight, so the ratio is about 1.0. For most jet aircraft, the landing is typically calculated at a weight of about 0.85 times the takeoff weight. Military design requirements will frequently specify full payload and some percent of fuel remaining (usually 50%) for the landing.

Arrested Landing

Aircraft which land on Navy aircraft carriers are stopped by a cable-and-brake arrangement called "arresting gear." One of several cables strung across the flight deck is caught by a hook attached to the rear of the aircraft. The cable is attached at both ends to drum mechanisms which exert a drag upon the cable as it is pulled by the aircraft, thus stopping it in a very short distance.

For carrier-based aircraft, the approach speed (1.15 times the stall speed) is the same as the touchdown speed. Carrier pilots do not flare and slow down for landing. Instead, they are taught to fly the aircraft right into the deck, relying upon the arresting gear to make the landing. By using this technique, the aircraft has enough speed to go around if the cables are missed.

The landing weight limits for three standard arresting gears are depicted in Fig. 5.6. This figure can be used to determine the allowable approach speed based upon a first-guess of the landing weight. The approach speed divided by 1.15 defines the stall speed, which can then be used to estimate the wing loading.

Wing Loading for Cruise

At this point we must bring in the use of two aerodynamic coefficients, C_{D_0} and “ e .” C_{D_0} is the zero lift drag coefficient, and equals approximately 0.015 for a jet aircraft, 0.02 for a clean propeller aircraft, and 0.03 for a dirty, fixed-gear propeller aircraft. The Oswald efficiency factor e is a measure of drag due to lift efficiency, and equals approximately 0.6 for a fighter and 0.8 for other aircraft. These coefficients are extensively discussed in Chapter 12. Chapter 12 also contains methods for estimation of C_{D_0} and e .

To maximize range during cruise, the wing loading should be selected to provide a high L/D at the cruise conditions. The following discussion provides methods for selecting wing loading to optimize cruise range.

A propeller aircraft, which loses thrust efficiency as speed goes up, gets the maximum range when flying at the speed for best L/D , while a jet aircraft maximizes range at a somewhat higher speed where the L/D is slightly reduced. The speed for best L/D can be shown to result in parasite drag equaling the induced drag (see Chapter 17). Therefore, to maximize

range a propeller aircraft should fly such that:

$$qSC_{D_0} = qS \frac{C_L^2}{\pi A e} \quad (5.12)$$

During cruise, the lift equals the weight, so the lift coefficient equals the wing loading divided by the dynamic pressure. Substitution into Eq. (5.12) allows solution for the required wing loading to maximize L/D for a given flight condition. This result [Eq. (5.13)] is the wing loading for maximum range for a propeller aircraft.

$$\text{Maximum Prop Range: } W/S = q\sqrt{\pi A e C_{D_0}} \quad (5.13)$$

As the aircraft cruises, its weight reduces due to the fuel burned, so the wing loading also reduces during cruise. Optimizing the cruise efficiency while the wing loading is steadily declining requires reducing the dynamic pressure by the same percent [see Eq. (5.13)]. This can be done by reducing velocity, which is undesirable, or by climbing to obtain a lower air density. This range optimizing technique is known as a “cruise-climb.”

A jet aircraft flying a cruise-climb will obtain maximum range by flying at a wing loading such that the parasite drag is three times the induced drag (see Chapter 12 for the derivation of this relationship). This yields the following formula for wing-loading selection for range optimization of jet aircraft.

$$\text{Maximum Jet Range: } W/S = q\sqrt{\pi A e C_{D_0}/3} \quad (5.14)$$

Frequently an aircraft will not be allowed to use the cruise-climb technique to maximize range. Air Traffic Controllers prefer that aircraft maintain a single assigned altitude until given permission to climb or descend to another altitude. The pilot will attempt to obtain permission from the Air Traffic Controllers to climb several times during the flight, thus forming the characteristic “stairstep climb schedule.” This allows the thrust setting to be maintained approximately at the setting that minimizes fuel consumption.

Wing Loading for Loiter Endurance

Most aircraft will have some loiter requirement during the mission, typically 20 min of loiter before landing. Unless the loiter requirement is a substantial fraction of the total mission duration, it is better to optimize the wing loading for cruise.

Patrol aircraft such as the ASW design example of Chapter 3 are sometimes more concerned with time on station than with cruise efficiency. Other aircraft which may be concerned with loiter endurance are airborne command posts and intelligence-gathering aircraft.

For an aircraft which must be optimized for loiter, the wing loading should be selected to provide a high L/D . For jet aircraft, the best loiter occurs at maximum L/D so Eq. (5.12) [repeated below as Eq. (5.15)] should be used. For a propeller aircraft, loiter is optimized when the induced drag is three times the parasite drag, which yields Eq. (5.16). This also provides the wing loading for minimum power required.

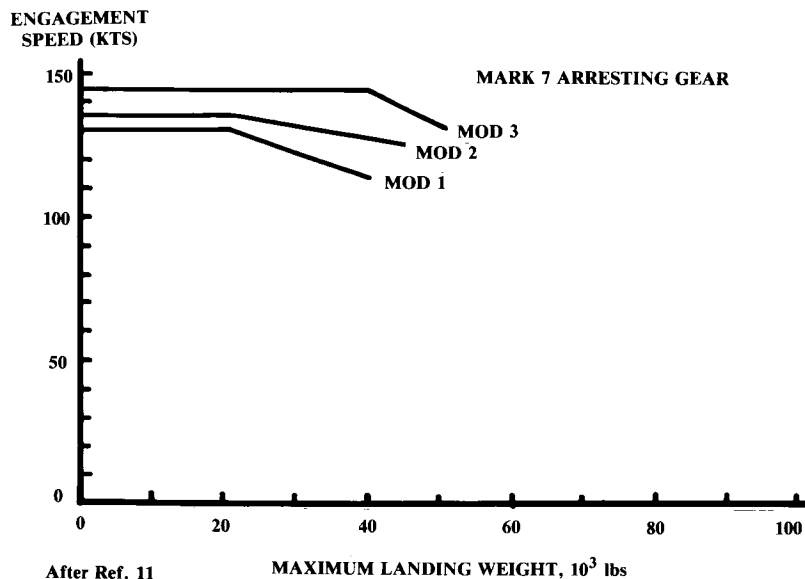


Fig. 5.6 Arresting gear weight limits.

$$\text{Maximum Jet Loiter: } W/S = q\sqrt{\pi A e C_{D_0}} \quad (5.15)$$

$$\text{Maximum Prop Loiter: } W/S = q\sqrt{3\pi A e C_{D_0}} \quad (5.16)$$

These equations assume that the loiter velocity and altitude are known. If the loiter altitude is not specified, it should be selected for best specific fuel consumption at the loiter power setting. This is typically 30,000–40,000 ft for a jet, and the limit altitude for the turbocharger for a piston-propeller aircraft. For a nonturbocharged engine, best loiter occurs at sea level.

Usually, the loiter velocity is not specified. Instead the designer must determine the best loiter velocity and select the wing loading accordingly. This requires cross plotting of wing loadings with the resulting L/D and specific fuel consumption for various velocities and altitudes. Such a procedure is too complex for initial design purposes.

For initial design purposes, it can be assumed that the best loiter velocity will be about 150–200 knots for turboprops and jets, and about 80–120 knots for piston-props. If altitude is not specified, the altitude for best fuel consumption should be selected.

The wing loading estimated from Eqs. (5.15) or (5.16) is the average during the loiter. This must be converted to takeoff conditions by dividing the loiter wing loading by the ratio of the average loiter weight to the takeoff weight. In the absence of better information, this ratio can be assumed to be about 0.85.

Remember that Eqs. (5.15) and (5.16) are to be used for designing an aircraft optimized solely for loiter. Optimizing for loiter alone is very rare in aircraft design. For most aircraft, the wing loading will be selected for best cruise and the loiter capabilities will be a “fallout.”

Instantaneous Turn

An aircraft designed for air-to-air dogfighting must be capable of high turn rate. This parameter, $d\psi/dt$ or $\dot{\psi}$, will determine the outcome of a dogfight if the aircraft and pilots are evenly matched otherwise. When air-to-air missiles are in use, the first aircraft to turn towards the other aircraft enough to launch a missile will probably win. In a guns-only dogfight, the aircraft with the higher turn rate will be able to maneuver behind the other. A turn rate superiority of 2 deg/s is considered significant.

There are two important turn rates. The “sustained” turn rate for some flight condition is the turn rate at which the thrust of the aircraft is just sufficient to maintain velocity and altitude in the turn. If the thrust acts approximately opposite to the flight direction, then the thrust must equal the drag for a sustained turn.

If the aircraft turns at a quicker rate, the drag becomes greater than the available thrust, so the aircraft begins to slow down or lose altitude. The “instantaneous” turn rate is the highest turn rate possible, ignoring the fact that the aircraft will slow down or lose altitude.

The “load factor,” or “ g -loading,” during a turn is the acceleration due to lift expressed as a multiple of the standard acceleration due to gravity ($g = 32.2 \text{ ft/s}^2$). Load factor (“ n ”) is equal to the lift divided by the aircraft’s weight.

Level, unturning flight implies a load factor of one ($n = 1$). In a level turn, the wing must provide 1- g lift in the vertical direction to hold up the aircraft, so the remaining “ g ’s” available to turn the aircraft in the horizontal direction are equal to the square root of n squared minus 1 (see Fig. 17.4). Thus the radial acceleration in a level turn is g times the square root of ($n^2 - 1$).

Turn rate is equal to the radial acceleration divided by the velocity. For a level turn, this results in Eq. (5.17). Note that this equation provides turn rate in radians per second, which must be multiplied by 57.3 to obtain degrees per second.

$$\dot{\psi} = \frac{g\sqrt{n^2 - 1}}{V} \quad (5.17)$$

where

$$n = \frac{qC_L}{W/S} \quad (5.18)$$

Instantaneous turn rate is limited only by the usable maximum lift, up to the speed at which the maximum lift exceeds the load-carrying capability of the wing structure. Typically, a fighter aircraft will be designed to an operational maximum load factor of 7.33 g , although newer fighters are being designed to 8- or 9- g . This g limit must be met at some specified combat weight.

The speed at which the maximum lift available exactly equals the allowable load factor is called the “corner speed,” and provides the maximum turn rate for that aircraft at that altitude. In a dogfight, pilots try to get to corner speed as quickly as possible as it provides the best turn rate. Typically, a modern fighter has a corner speed of about 300–350 knots indicated airspeed (i.e., dynamic pressure) regardless of altitude.

Design specifications will usually require some maximum turn rate at some flight condition. Equation (5.17) can be solved for the load factor at the specified turn rate as follows:

$$n = \sqrt{\left(\frac{\dot{\psi}V}{g}\right)^2 + 1} \quad (5.19)$$

If this value of load factor is greater than the ultimate load factor specified in the design requirements, somebody has made a mistake. The required wing loading can be solved for in Eq. (5.18) as follows:

$$\frac{W}{S} = \frac{qC_{L_{\max}}}{n} \quad (5.20)$$

The only unknown is the maximum lift coefficient at combat conditions. This is not the same as the maximum lift coefficient for landing. During combat, use of full flap settings is not usually possible. Also, there is a Mach-number effect which reduces maximum lift at higher speeds. Fre-

quently the combat maximum usable lift will be limited by buffeting or controllability considerations.

For initial design purposes, a combat maximum lift coefficient of about 0.6–0.8 should be assumed for a fighter with only a simple trailing-edge flap for combat. For a fighter with a complex system of leading- and trailing-edge flaps which can be deployed during combat, a maximum usable lift coefficient of about 1.0–1.5 is attainable. Chapter 12 provides better methods of estimating the maximum lift coefficient.

Again, the resulting wing loading must be divided by the ratio of combat weight to takeoff weight to obtain the required takeoff wing loading. Usually the combat weight is specified as the aircraft design takeoff weight with any external fuel tanks dropped and 50% of the internal fuel gone. This is approximately 0.85 times the takeoff weight for most fighters.

The resulting wing loading is the maximum which will allow the required instantaneous turn.

Sustained Turn

The sustained turn rate is also important for success in combat. If two aircraft pass each other in opposite directions, it will take them about 10 seconds to complete 180-deg turns back towards the other. The aircraft will probably not be able to maintain speed while turning at the maximum instantaneous rate. If one of the aircraft slows down below corner speed during this time it will be at a turn rate disadvantage to the other, which could prove fatal.

Sustained turn rate is usually expressed in terms of the maximum load factor at some flight condition that the aircraft can sustain without slowing or losing altitude. For example, the ability for sustaining 4- or 5-g at 0.9 Mach number at 30,000 ft is frequently specified. Equations (5.17) or (5.19) can be used to relate turn rate to load factor.

If speed is to be maintained, the thrust must equal the drag (assuming that the thrust axis is approximately aligned with the flight direction). The lift must equal the weight times the load factor, so we can write:

$$n = (T/W) (L/D) \quad (5.21)$$

Sustained—turn load factor is maximized by maximizing the T/W and L/D . The highest L/D occurs when the induced drag equals the parasite drag, as expressed by Eq. (5.12). During a turn, the lift equals the weight times n , so the lift coefficient equals the wing loading times n divided by the dynamic pressure. Substitution into Eq. (5.12) yields:

$$W/S = \frac{q}{n} \sqrt{\pi A e C_{D0}} \quad (5.22)$$

This equation gives the wing loading that maximizes the sustained turn rate at a given flight condition. Note that if n equals one, Eq. (5.22) is the same as Eq. (5.13), the wing loading for best L/D in level flight.

Equation (5.22) estimates the wing loading that maximizes the sustained turn rate regardless of thrust available. This equation will frequently give

ridiculously low values of wing loading that will provide the required sustained turn rate using only a fraction of the available thrust.

The wing loading to exactly attain a required sustained load factor n using all of the available thrust can be determined by equating the thrust and drag, and using the fact that since lift equals weight times n , the lift coefficient during maneuver equals the wing loading times n , divided by the dynamic pressure. This yields Eq. (5.23).

$$T = q S C_{D0} + q S \left(\frac{C_L^2}{\pi A e} \right) = q S C_{D0} + \frac{n^2 W^2}{q S \pi A e} \quad (5.23)$$

or

$$\frac{T}{W} = \frac{q C_{D0}}{W/S} + \frac{W}{S} \left(\frac{n^2}{q \pi A e} \right) \quad (5.24)$$

Equation (5.24) can be solved for W/S to yield the wing loading that exactly attains a required sustained load factor n [Eq. (5.25)]. Also, Eq. (5.24) can be used later to recheck the T/W after the wing loading is selected.

$$\frac{W}{S} = \frac{(T/W) \pm \sqrt{(T/W)^2 - (4n^2 C_{D0}/\pi A e)}}{2n^2/q \pi A e} \quad (5.25)$$

The thrust-to-weight ratio for this calculation is at combat conditions, so the takeoff T/W must be adjusted to combat conditions by dividing by the ratio between combat and takeoff weight, and by multiplying by the ratio between combat thrust and takeoff thrust.

If the term within the square root in Eq. (5.25) becomes negative, there is no solution. This implies that, at a given load factor, the following must be satisfied regardless of the wing loading:

$$\frac{T}{W} \geq 2n \sqrt{\frac{C_{D0}}{\pi A e}} \quad (5.26)$$

It is very important to realize in these calculations that the efficiency factor e , is itself a function of the lift coefficient at which the aircraft is operating. This is due to the separation effects at higher lift coefficients that increase drag above the parabolic drag polar values. At high angles of attack the e value may be reduced by 30% or more.

Unfortunately, the above equations for turning flight are very sensitive to the e value. If these equations yield W/S values far from historical values, the e value is probably unrealistic and the calculated W/S values should be ignored. Methods in Chapter 12 will better account for the separation effects.

Climb and Glide

Appendix A.5 cites numerous climb requirements for FAR or military aircraft. These specify rate of climb for various combinations of factors

such as engine-out, landing-gear position, and flap settings. While the details may vary, the method for selecting a wing loading to satisfy such requirements is the same.

Rate of climb is a vertical velocity, typically expressed in feet-per-minute (which must be converted to feet-per-second for the following calculations). Climb gradient, " G ," is the ratio between vertical and horizontal distance traveled. As will be shown in Chapter 17, at normal climb angles the climb gradient equals the excess thrust divided by the weight, i.e.—

$$G = (T - D)/W \quad (5.27)$$

or

$$\frac{D}{W} = \frac{T}{W} - G \quad (5.28)$$

D/W can also be expressed as in Eq. (5.29), where in the final expression the lift coefficient is replaced by (W/qS) .

$$\frac{D}{W} = \frac{qSC_{D0} + qS(C_L^2/\pi Ae)}{W} = \frac{qC_{D0}}{W/S} + \frac{W}{S} \frac{1}{q\pi Ae} \quad (5.29)$$

Equating Eqs. (5.28) with (5.29) and solving for wing loading yields:

$$\frac{W}{S} = \frac{[(T/W) - G] \pm \sqrt{[(T/W) - G]^2 - (4C_{D0}/\pi Ae)}}{2/q\pi Ae} \quad (5.30)$$

Note the similarity to Eq. (5.25). Equation (5.30) is merely Eq. (5.25) for a load factor of 1.0, with the (T/W) term replaced by $[(T/W) - G]$. As before, T/W must be ratioed to the flight conditions and weight under consideration. The resulting W/S must then be ratioed to a takeoff-weight value.

The term within the square root symbol in Eq. (5.30) cannot go below zero, so the following must be true regardless of the wing loading:

$$\frac{T}{W} \geq G + 2 \sqrt{\frac{C_{D0}}{\pi Ae}} \quad (5.31)$$

This equation says that no matter how "clean" your design is, the T/W must be greater than the desired climb gradient! [T/W for a propeller aircraft was defined in Eq. (5.1).]

Another implication of this equation is that a very "clean" aircraft that cruises at a high speed despite a very low T/W will probably climb poorly. A 200-mph airplane that flies on 20 hp can't be expected to climb as well as an airplane that requires 200 hp to reach 200 mph (unless the latter weighs ten times as much).

C_{D0} and e values for some of the climb conditions specified in Appendix A.5 must include the effects of flaps and landing gear. Chapter 12 will

provide methods for estimating these effects; but for now, approximations can be used.

For takeoff flap settings, C_{D0} will increase by about 0.02 and e will decrease about 5%. For landing flap settings, C_{D0} will increase by about 0.07 and e will decrease by about 10% relative to the no-flap value. Retractable landing gear in the down position will increase C_{D0} by about 0.02 (Ref. 7).

Sometimes the rate of climb must also be calculated with one engine windmilling or stopped. The thrust loss due to a "dead" engine can be accounted for in the T/W . For example, if a three-engined aircraft loses one engine, the T/W becomes two-thirds of the original T/W .

The drag increase due to a windmilling or stopped engine will further reduce the climb rate. Chapter 12 provides methods for estimating this drag. For rough initial analysis, however, it can probably be ignored.

Equation (5.30) can also be used to establish the wing loading required to attain some specified glide angle, by setting T/W to zero and using a negative value of G (i.e., a glide in the negative direction). If a particular sink rate must be attained, the value of G to use is the sink rate divided by the forward velocity. Make sure that both are in the same units.

Maximum Ceiling

Equation (5.30) can be used to calculate the wing loading to attain some maximum ceiling, given the T/W at those conditions. The climb gradient G can be set to zero to represent level flight at the desired altitude. Frequently a small residual climb capability, such as 100 ft/min, is required at maximum ceiling. This can be included in Eq. (5.30) by first solving for the climb gradient G (climb rate divided by forward velocity).

For a high-altitude aircraft such as an atmospheric research or reconnaissance plane, the low dynamic pressure available may determine the minimum possible wing loading. For example, at 100,000 ft and 0.8 Mach number, the dynamic pressure is only 10 psi. Equation (5.13) [repeated below as Eq. (5.32)] can be used to determine the wing loading for minimum power.

$$W/S = q\sqrt{\pi AeC_{D0}} \quad (5.32)$$

This may suggest a wing loading so low as to be impractical, and so should be compared with the wing loading required to fly at a given lift coefficient, i.e.:

$$W/S = qC_L \quad (5.33)$$

For efficiency during high-altitude cruise, the lift coefficient should be near the airfoil design lift coefficient. For a typical airfoil, this is about 0.5. For a high-altitude aircraft, new high-lift airfoils with design lift coefficients on the order of 1.0–1.4 can be used.

5.4 SELECTION OF THRUST-TO-WEIGHT AND WING LOADING

An initial estimate of the thrust-to-weight (or horsepower-to-weight) ratio was previously made. From the wing loadings estimated above, the lowest

value should be selected to ensure that the wing is large enough for all flight conditions. Don't forget to convert all wing loadings to takeoff conditions prior to comparisons.

A low wing-loading will always increase aircraft weight and cost. If a very low wing-loading is driven by only one of the requirements, a change in design assumptions (such as a better high-lift system) may allow a higher wing-loading.

Also, keep in mind that the wing loadings calculated by Eqs. (5.13–5.16), and Eqs. (5.22) and (5.32) are aerodynamic optimizations for only a portion of the mission. If these give wing loadings far lower than those in Table 5.5, they may be ignored.

When the best compromise for wing loading has been selected, the thrust-to-weight ratio should be rechecked to ensure that all requirements are still met. The equations in the last section which use T/W should be recalculated with the selected W/S and T/W . Only then can the next step of design, initial sizing, be initiated.

INITIAL SIZING

6.1 INTRODUCTION

Aircraft sizing is the process of determining the takeoff gross weight and fuel weight required for an aircraft concept to perform its design mission. Sizing was introduced in Chapter 3, in which a quick method based upon minimal information about the design was used to estimate the sizing parameters. That sizing method was limited to fairly simple design missions. This chapter presents a more refined method capable of dealing with most types of aircraft-sizing problems.

An aircraft can be sized using some existing engine or a new design engine. The existing engine is fixed in size and thrust, and is referred to as a “fixed-engine” (“fixed” refers to engine size).

The new design engine can be built in any size and thrust required, and is called a “rubber engine” because it can be “stretched” during the sizing process to provide any required amount of thrust.

Rubber-engine sizing is used during the early stages of an aircraft development program that is sufficiently important to warrant the development of an all-new engine. This is generally the case for a major military fighter or bomber program, and is sometimes the case for a transport-aircraft project such as the SST.

In these cases, the designer will use a rubber engine in the early stages of design, and then, with the customer, tell the engine company what characteristics the new engine should have. When the engine company finalizes the design for the new engine, it becomes fixed in size and thrust. The aircraft concept will then be finalized around this now-fixed engine.

Developing a new jet engine costs on the order of a billion dollars. Developing and certifying a new piston engine is also very expensive. Most aircraft projects do not rate development of a new engine, and so must rely on selecting the best of the existing engines. However, even projects which must use an existing engine may begin with a rubber-engine design study to determine what characteristics to look for in the selection of an existing engine.

The rubber engine can be scaled to any thrust so the thrust-to-weight ratio can be held to some desired value even as the aircraft weight is varied. The rubber-engine sizing approach allows the designer to size the aircraft to meet both performance and range goals, by solving for takeoff gross weight while holding the thrust-to-weight ratio required to meet the performance objectives. As the weight varies, the rubber-engine is scaled up or down as required.